

LayerPack: Training-Free Animation Inbetweening with Cross-Layer Consistency

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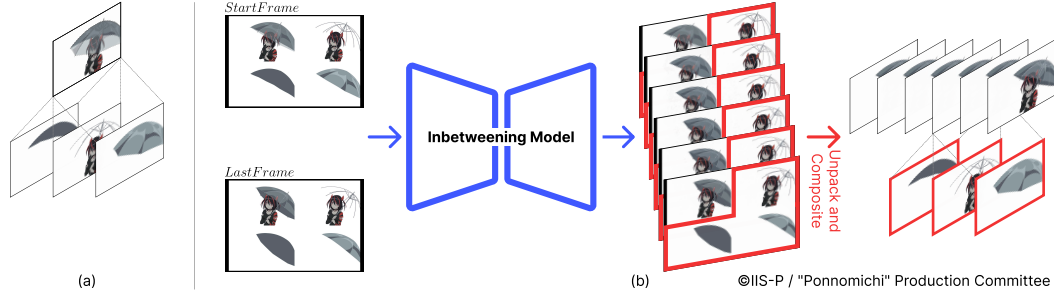


Figure 1: (a) Separated layers as commonly produced in 2D hand-drawn animation. (b) Proposed LayerPack method, where multiple layers are packed into a single image, referred to as a LayerPack image. The input comprises two LayerPack images, i.e., the start and end frames, where each LayerPack image includes both the layers and the composited image. A diffusion model performs inbetweening to generate multiple intermediate frames. Finally, the LayerPack images are unpacked into layers, and we obtain the layer-separated video.

ACM Reference Format:

Atsushi Maruyama, Sosui Koga, Akinobu Maejima, Yuki Koyama, I-Chao Shen, and Takeo Igarashi. 2025. LayerPack: Training-Free Animation Inbetweening with Cross-Layer Consistency. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771440>

1 Introduction

In animation production, after drawing keyframes that capture the extremes of motion, considerable labor is required for inbetweening, which smoothly interpolates the motion between two keyframes. Recently, diffusion models for inbetweening have been studied actively, and impressive results have been demonstrated [Peebles and Xie 2023; Xing et al. 2024; Zhang and Agrawala 2025].

Layer-separated data, as shown in Figure 1(a), are widely used in professional animation workflows. By compositing these layers

with alpha blending, the richness of expression in animation can be enhanced significantly. Consequently, inbetweening is performed for each layer individually, and it is crucial that the generated layers remain spatially synchronized after compositing. Here, key inbetweening must be performed for each layer, and the resulting layers must remain spatially synchronized after compositing.

However, performing inbetweening with diffusion models while preserving the layer separation is challenging for three main reasons. First, handling the transparency dimension (i.e., the alpha channel) is inherently difficult. Second, datasets containing multilayer structures are scarce. LayerAnimate [Yang et al. 2025] attempted to create a dataset of layers; however, it differs from the data used in professional animation workflows. Third, the number of layers varies depending on the scene and the frame. Moreover, when generating each layer independently, the resulting motions of the layers frequently do not remain synchronized even when denoising from identical noise. LayerAnimate [Yang et al. 2025] was designed to control each layer with different magnitudes of motion; however, it fails to maintain cross-layer consistency.

To address this issue, we propose LayerPack, which is a training-free method for diffusion models for multilayer inbetweening that ensures synchronized motion across layers. The proposed LayerPack method packs content from multiple layers into a grid and

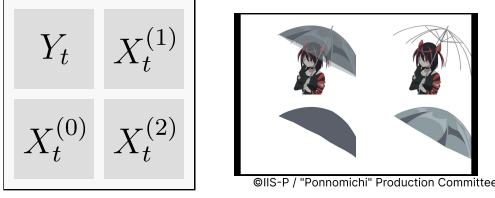


Figure 2: The composited image and individual layer images are packed into a single image. For example, for a sequence with three layers, the four images (i.e., three layers and one composited image) are arranged in a 2×2 grid.

sends it to a standard video generation model without requiring additional training processes. The results of qualitative observations and quantitative evaluations demonstrate that the proposed LayerPack method generates higher-quality outputs than generating a video for each layer independently.

2 Method

LayerPack is a training-free method to generate multilayer inbetweening with cross-layer consistency. [Section 2.1](#) defines the target problem, and [Section 2.2](#) describes the layerwise frame representation implemented in the proposed LayerPack method.

2.1 Problem Definition

In this study, we focus on the *layer-wise inbetweening* task. Here, the input comprises a start frame and an end frame, which are separated into layers. The goal is to generate intermediate frames for each layer and composite them to create the inbetweening frames. Let the total number of animation frames be $T + 1$ and the number of layers be L , where the frame index is $t \in \{0, 1, 2, \dots, T\}$. The layer index l is ordered such that smaller and larger values correspond to the background and the foreground, respectively.

The image of layer l in frame t is denoted $X_t^{(l)} \in \mathbb{R}^{3 \times H \times W}$. The input comprises all layer images for the keyframes $t = 0$ and $t = T$, i.e., $\{X_t^{(l)} \mid t \in \{0, T\}, l \in \{0, 1, \dots, L - 1\}\}$. The output is the set of all layer images for all frames: $\{X_t^{(l)} \mid t \in \{0, 1, \dots, T\}, l \in \{0, 1, \dots, L - 1\}\}$. For compositing, alpha blending is applied in ascending order of the layer indices, and the alpha compositing operation is given as $Y_t = C(X_t^{(0)}, \dots, X_t^{(L-1)})$.

2.2 Layer-wise Frame Representation

In the proposed LayerPack method, the input and output images are packed into a grid. Here, the packed image is referred to as a *LayerPack image* ([Figure 2](#)) and is defined as follows.

The LayerPack image P_t for frame t is constructed by spatially concatenating all layer images for frame t along with the composited image obtained by alpha blending. In other words, P_t contains the L layer images $\{X_t^{(l)} \mid l \in \{0, 1, \dots, L - 1\}\}$ and the composited image Y_t , resulting in a total of $L + 1$ images. These images are arranged on a grid with K rows and K' columns ($K' = K$ or $K + 1$), and each cell in the grid contains an image of size (H, W) . Thus, the entire LayerPack image P_t forms a single image of size $(K \times H, K' \times W)$.

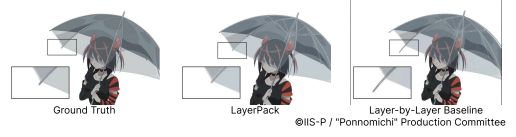


Figure 3: Comparison between the ground truth, the proposed LayerPack method, and the layer-by-layer baseline method. The figure shows the middle frame of the generated video (at 0.5 s). In the regions highlighted by boxes, the LayerPack method produces synchronized layers, whereas the layer-by-layer baseline method does not. Furthermore, the result of the proposed method is more similar to the ground truth frame.

Table 1: Quantitative comparison. We performed a quantitative evaluation on 10 different scenes. The proposed LayerPack method significantly outperformed the baseline methods.

method	PSNR ($\uparrow$)	LPIPS ($\downarrow$)
LayerPack (Ours)	10.73	0.053
Baseline (Layer-by-Layer)	6.07	0.135

Using P_0 and P_T as the inputs to a diffusion transformer-based inbetweening model, we generate the set $\{P_t\}_{t=0}^T$.

3 Experiments and Results

We evaluated the proposed LayerPack method on 1-s anime-style videos that were manually separated into two or three layers by an artist. As a baseline method for comparison, we used a method that performs inbetweening independently for each layer using the L pairs of keyframes $\{(X_0^{(l)}, X_T^{(l)})\}$ for $l \in \{0, 1, \dots, L - 1\}$. Note that both methods employ FramePack [[Zhang and Agrawala 2025](#)] as an image-to-video diffusion model.

In this experiment three types of evaluations were performed, i.e., a quantitative evaluation of the layer-separated results, a quantitative evaluation after compositing, and a qualitative evaluation. For the quantitative evaluations, we used the peak signal-to-noise ratio (PSNR) and the learned perceptual image patch similarity (LPIPS) as metrics for 10 generated results ([Table 1](#)). The results of the qualitative evaluation are shown in [Figure 3](#). As highlighted in the boxes in these figures, the proposed LayerPack method produced synchronized motion between independently generated layers, particularly around object boundaries. In addition, temporal and spatial consistency was achieved (refer to the supplementary video).

4 Conclusion and Discussion

We have proposed the *LayerPack* method, which is a training-free layer-wise frame representation method for multilayer inbetweening that achieves synchronized motion across layers. The experimental results demonstrated that even when a single object is separated into multiple layers, the generated motions remain temporally and spatially synchronized.

Acknowledgments

This work is funded by New Energy and Industrial Technology Development Organization (NEDO) JPNP20017.

References

- William Peebles and Saining Xie. 2023. Scalable Diffusion Models with Transformers. arXiv:2212.09748 [cs.CV] <https://arxiv.org/abs/2212.09748>
- Jinbo Xing, Hanyuan Liu, Menghan Xia, Yong Zhang, Xintao Wang, Ying Shan, and Tien-Tsin Wong. 2024. Toonrafter: Generative cartoon interpolation. *ACM Transactions on Graphics (TOG)* 43, 6 (2024), 1–11.
- Yuxue Yang, Lue Fan, Zuzeng Lin, Feng Wang, and Zhaoxiang Zhang. 2025. LayerAnimate: Layer-level Control for Animation. arXiv:2501.08295 [cs.CV] <https://arxiv.org/abs/2501.08295>
- Lymin Zhang and Maneesh Agrawala. 2025. Packing Input Frame Context in Next-Frame Prediction Models for Video Generation. arXiv:2504.12626 [cs.CV] <https://arxiv.org/abs/2504.12626>